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RydFixr: A Trust-Centric Digital Platform for Transparent and Secure Bike Repair and Mobility Assistance

K Guru Dheeraj Reddy
Department of CSE(Data Science)
Presidency University
Bengaluru, India
dheeraj.20221csd0099@presidencyuniversity.in

Rohith D Raj
Department of CSE(Data Science)
Presidency University
Bengaluru,India
rohith.20221csd0097@presidencyuniversity.in

Dr.Wasim Yasin
School of Computer Science and Engineering
Presidency University
Bengaluru, India
wasim@presidencyuniversity.in

***Abstract*— in many real-world scenarios, bike repair services lack transparency, accountability, and structured workflows, often leading to customer dissatisfaction and disputes. This paper presents *rydFixr*, a trust-driven digital platform designed to bridge the gap between bike owners and verified service providers through a centralized system.**

The propose solution provides for a fair pricing system by keeping labor and spare parts expenses separate, guaranteeing that technicians earn money while allowing clients to know all their costs. The solution provides for an escrow system that prevents misuse, releasing money to the right individuals only when services have been proven. There will be proof-based service validation, warranty based on rework only, and third-party inspection. Security will also be enhanced with a safe ride system that monitors the journey in real time. The technology used in designing the system is up-to-date web technology, and there is a modular and role-based design.

Keywords—Bike Repair Platform, Escrow Payment, Service Transparency, Digital Marketplace, Trust-Based System, Mobility Assistance

I. INTRODUCTION

Currently, in an ordinary urban or semi-urban setting, the services provided by bicycle repair shops have not been formalized. The customers end up getting their bikes repaired from roadside bike mechanics or bike garages. In such a setting, customers may not be able to get any assurance of the quality of the service and the price at which the work is being done. Moreover, once the repair work is done, the customer will have no record of the same, meaning there can be no guarantees and no follow-up on problems that arise. This results in lack of transparency and disconnect between the customer and the mechanic or shopkeeper. On one hand, the customer ends up doubting the price and the quality of the service; whereas, on the other hand, the mechanic or shopkeeper may experience irregular income and constant negotiating. This problem can be tackled via the use of rydFixr, which will emphasize trust, transparency, and structured execution.

Apart from the aforementioned constraints, another significant drawback of the current repair mechanisms lies in the lack of an organized workflow process where responsibility could be held at every stage of the service delivery process. Normally, when the repair work is done, there is no validation process in place to ensure that the problem has been sorted out successfully. Consequently, the customers continue facing such problems again and again and have to visit the same service agencies with no certainty of a solution being found. In cases where users undergo failures when using the systems in an unfamiliar or isolated environment, it becomes impossible to monitor the user's welfare. This aspect raises an important issue of lack of a safe way within existing mobility aid systems. As more digital services become popular with consumers on a daily basis, it becomes necessary to adopt more efficient systems that ensure security, accountability, and reliability. Any efficient platform has to meet the needs of both customers and service providers while being sustainable. This problem can be solved by rydFixr which represents an efficient framework for developing trustworthy and effective systems by including secure workflows, financial transactions, and verification of service delivery.

II. LITERATURE REVIEW

The development of the proposed rydFixr system will be influenced by the extensive literature review that was undertaken to analyze the existing literature on the topic of vehicle breakdown services and assistance and management systems. As found by previous researchers, vehicle breakdowns are rather frequent phenomena and can cause major inconveniences to consumers who tend to get into such situations, especially when the vehicle breaks down in an area which is not familiar to the user and no assistance is available nearby [4], [8]. There are various solutions that have been developed in order to establish connections between customers and their nearest available service providers. The use of GPS technology in order to track down the nearest service provider is an excellent way to accomplish such a task. Systems have already been proposed by the works presented in [2] and [3]. They show that the nearest Fixpert could easily be identified using this approach. We have used the same concept in order to build our system. However, in summary, the framework presented is not conceived de novo but is well-constructed on top of previous literature while taking into consideration their shortcomings. With the integration of location matching [2], [3], a systematic approach [6], [9], and effective communication techniques [8], rydFixr offers a more streamlined and user-oriented platform for bike repairs and mobility support.

III. METHODOLOGY

The methodology behind the suggested rydFixr system is developed in such a way to ensure the process-driven approach that will guarantee visibility, responsibility, and control of the process of providing bike repair services. Overall, the suggested system relies on the centralized back-end approach where all user interactions occur through the processes, not directly.

A. System Workflow Design

It starts with the lifecycle for the services, which is based on sequence starting with service request creation, all the way up to the payment and, possibly, some form of rework. At each step in the

flow, there is an established state defined by the system, including request creation, quote, payment, execution, and completion.

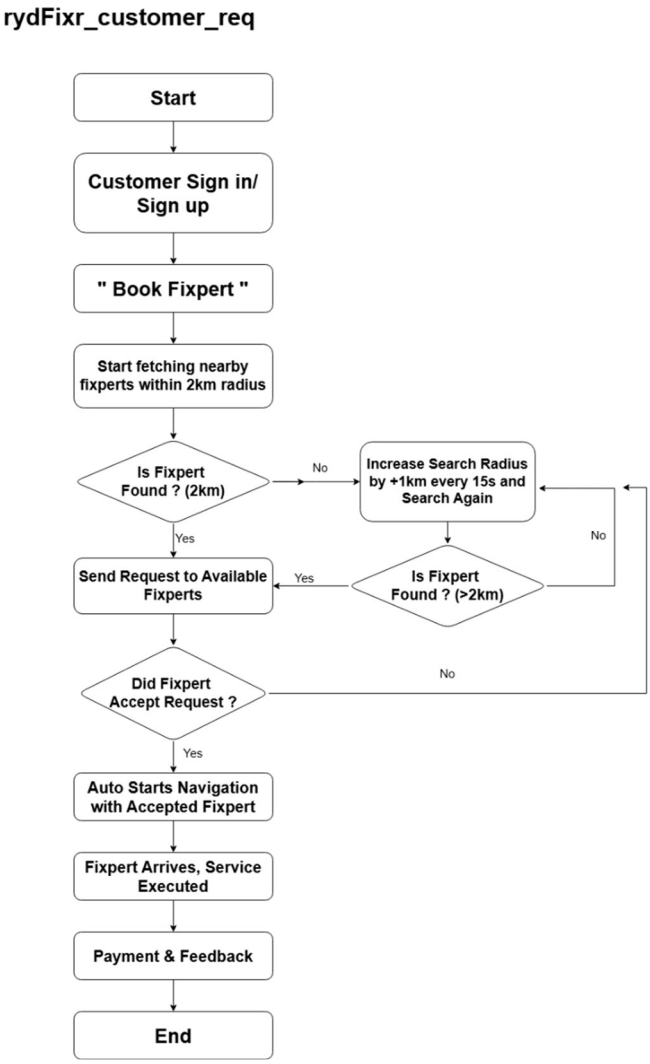

Figure. 1 rydFixr Customer Request

B. *Service Request and Fixpert Assignment*

It starts when a client makes a maintenance request by describing the problem type and its location. Subsequently, the system alerts Fixperts near the location depending on their availability.

As shown in the figure below, Fixperts have an opportunity to see client information, including location, bike problems, and distance, before accepting or rejecting the request. After accepting the request, the system locks the request to the Fixpert, which prevents multiple Fixperts from receiving the same request.

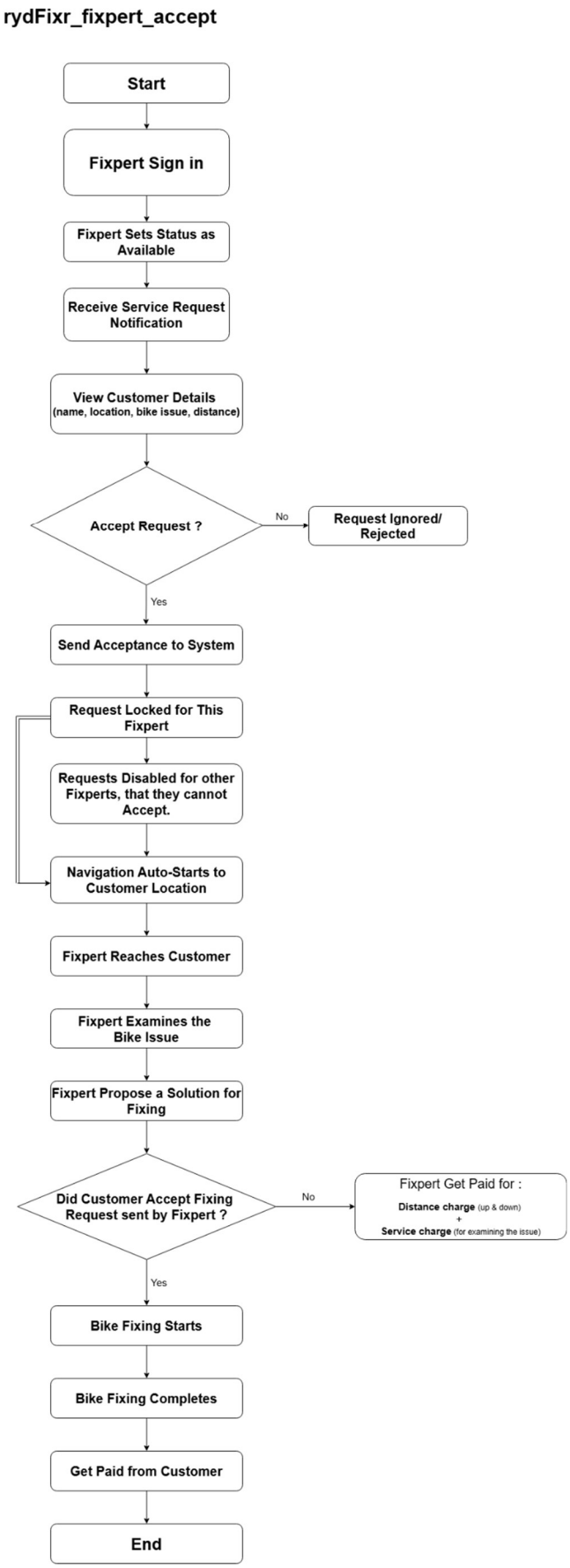

Figure. 2 rydFixr Fixpert Assignment

C. Inspection and Quotation Process

Once the problem with the bike has been inspected at the customer's site, the Fixpert will propose an appropriate solution to the problem. Pricing is divided into two segments; first, the price for labour and the second one is spare parts. The quoted prices are forwarded to the customer to confirm whether he accepts them or not. Failure to do so means that only inspection charges are paid to the Fixpert.

D. Escrow-Based Payment Mechanism

After the payment for the service has been verified, the process of repairing begins, conducted by the Fixpert. The service is provided in person at the customer's premises. As part of the process of repairing, the Fixpert should prove the performance of the repair process by sending photographs or videos as proof. Proof is used as a means of validating the performance of the service rendered to reduce ambiguity and establish trust between the two sides. However, when the repair requires high risks or complexity, an extra element of validation could be brought in the form of appointing a third party. This third party will act as an independent validator of the repairing process and ensure that it has been conducted properly while remaining impartial to both sides involved in the transaction. Once approved, the system proceeds to release the funds held in escrow. This process involves releasing the money into the accounts of Fixer and other parties according to the set structure. As such, the system links financial transactions to services delivered. For addressing any problems occurring after services have been delivered, the system adopts a rework-based warranty framework rather than providing refunds to the customers. Should a similar problem recur in a stipulated time frame, the customer can make a request for a rework. It is the responsibility of the same Fixer to solve the problem at hand without requiring extra charges. In addition to service delivery reliability, user safety has also been integrated into the system. Safe Ride is one of the features provided by the application to ensure safety during rides. The system tracks the user throughout the duration of a ride by sending regular check-up calls and escalating the matter when necessary. If a user does not respond to the call, the system will contact the administrative office in an attempt to reach the user. All these processes run from the backend system in the form of a centralized management platform.

E. Distance & Routing (Haversine Formula)

$$d = 2r * \text{asin}(\text{sqrt}(\sin^2((\varphi 2 - \varphi 1)/2) + \cos(\varphi 1) * \cos(\varphi 2) * \sin^2((\lambda 2 - \lambda 1)/2)))$$

Where:

d = total distance
φ = latitude
λ = longitude
r = Earth radius (~6371 km)

IV. IMPLEMENTATION

The development of the rydFixr application is done using a full-stack approach that concentrates on real-time communication, location-based matching, and management of the workflow process. The back-end of the project is coded using Node.js and Express.js, whereas the database uses MongoDB to store the user information, request details, and system status. Once a user makes a request regarding the problem at hand, the system saves this request together with the type of problem and geographical position into the database. The database is searched by all the Fixperts who are currently online and matched with those within the specified radius using a formula based on geographically positioned coordinates. Only the selected Fixperts receive the request in real-time using socket programming.

The acceptance mechanism of Fixpert is another integral component, ensuring that there is one-to-one relation of a request and Fixpert who accepted it. If a request gets accepted by a Fixpert, the availability of Fixpert changes to unavailable, and the request is locked from any further acceptances. Notifications regarding the acceptance of the request are sent to other Fixperts as well as customers, for keeping everyone on the same page. The assignment of Fixpert opens the way for tracking their real time location through the system, which will keep on sending continuous updates of locations shared by Fixperts that are further sent to customers. Moreover, distance from Fixpert and customers gets constantly updated in the system for keeping track during the service process. Event-driven approach is used by our system to handle various events including creating a request, acceptance of request, and updating of locations, and other real-time operations. As a result, real-time communication is ensured among all users without manual refreshing. All the tasks and processes are executed in the back end, which keeps record of every stage in order to maintain data consistency and error handling within each stage of lifecycle.

V.RESULTS AND DISCUSSION

It is evident that the use of rydFixr has been more effective compared to previous bike repair techniques due to the introduction of workflows and timely coordination. It was put through rigorous testing in various phases, including requesting services, assigning Fixpert tasks, tracking, and finally completing the service. Location-based matching makes it easier for the customer to be paired with a nearby Fixpert, making the process faster and more efficient. The algorithm ensures that only one Fixpert is assigned to complete the task.

The real-time monitoring system helps improve transparency, as customers can track the location of the Fixpert in addition to receiving dynamic distance updates. The system ensures that there is no more uncertainty on the part of customers and makes them feel comfortable especially when they need urgent repairs. By using the escrow system, transactions only take place once services are delivered and verified, thereby balancing the trust between customers and Fixperts. Similarly, the proof-based system helps improve accountability through the use of videos or pictures to confirm that the repair was done. Furthermore, the rework warranty system helps to ensure that all problems are fixed free of charge.

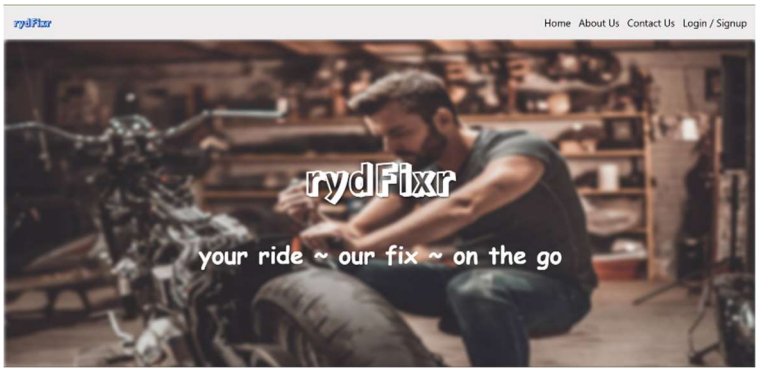

Figure. 3 rydFixr index page

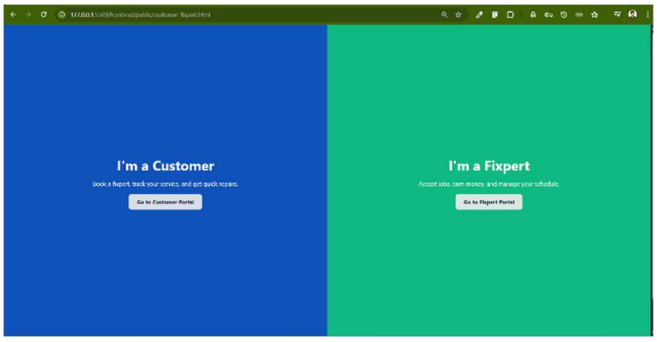

Figure. 4 rydFixr Customer-Fixpert Interface

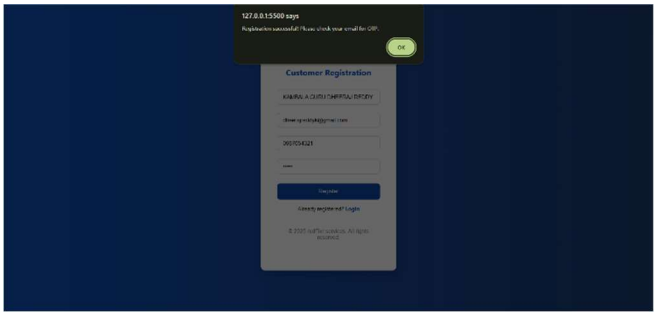

Figure. 5 rydFixr Customer Registration

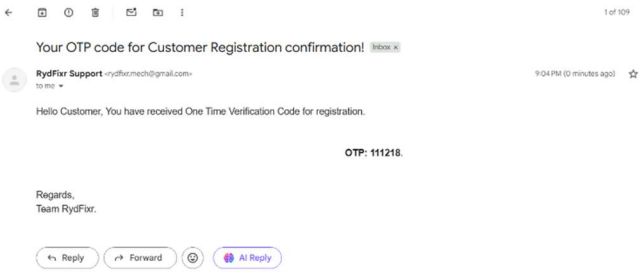

Figure. 6 rydFixr OTP

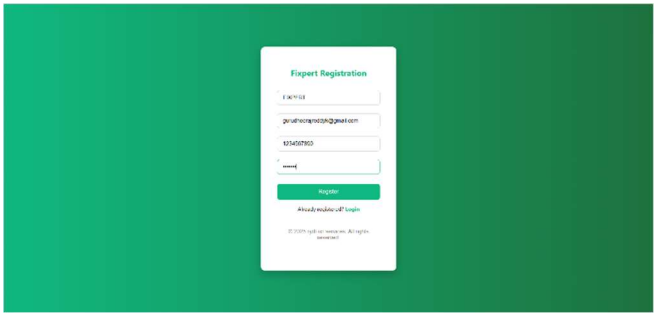

Figure. 7 rydFixr Fixpert Registration

Figure. 8 rydFixr Fixpert Login Credentials

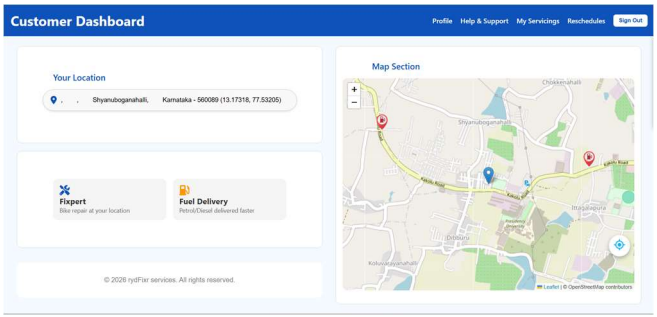

Figure. 9 rydFixr Customer-Dashboard + integrated Map

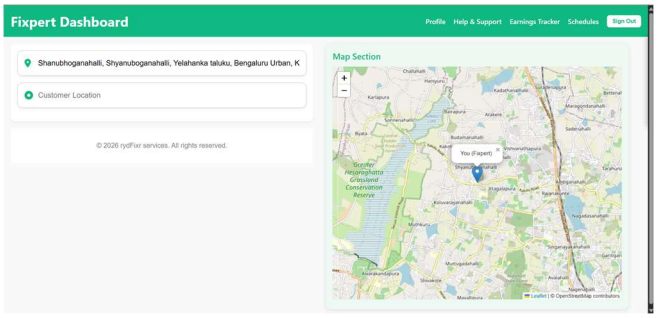

Figure. 10 rydFixr Fixpert-Dashboard + integrated Map

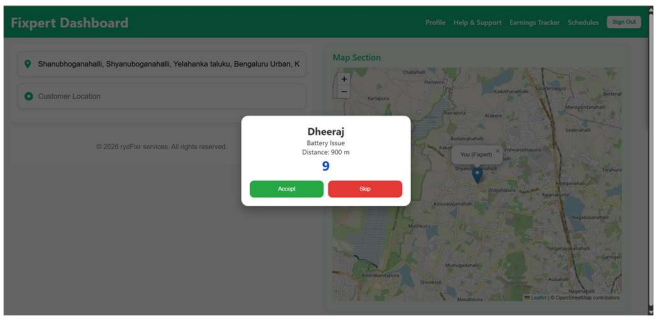

Figure. 11 rydFixr Fixpert service_request Acceptance

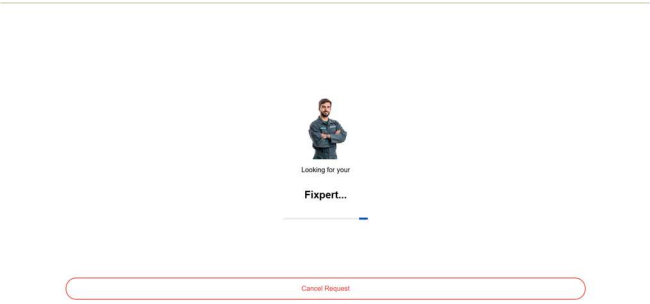

Figure. 12 rydFixr Searching Fixpert for Customer

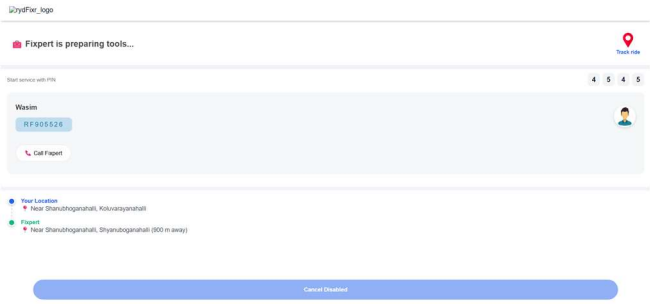

Fig. 13 rydFixr Accepted Fixpert details

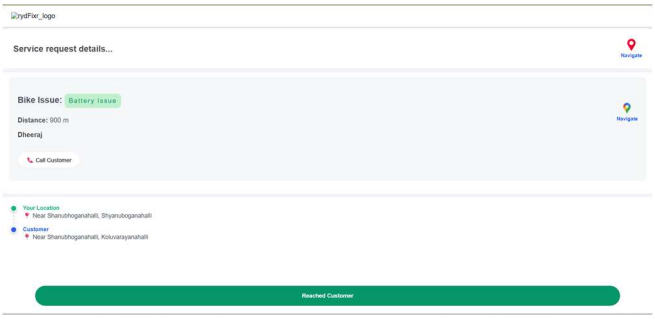

Figure. 14 rydFixr Service Request details

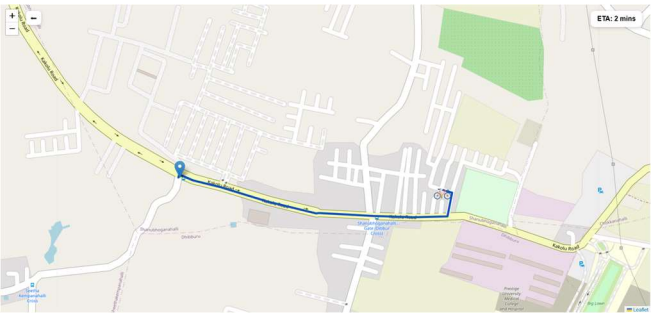

Figure. 15 rydFixr tracking fixpert page

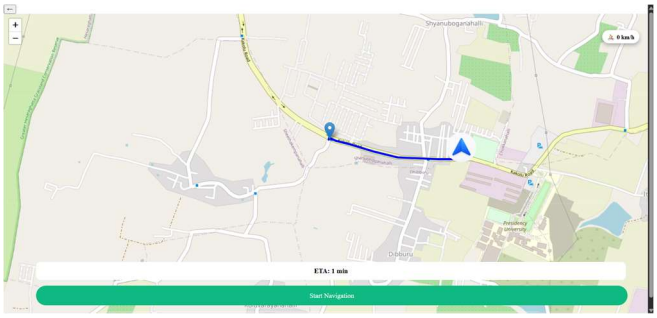

Figure. 16 rydFixr navigating to customer page

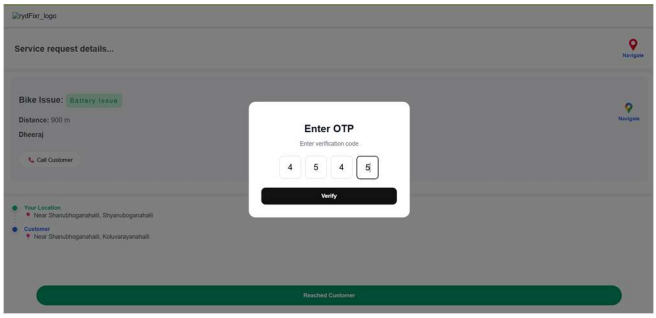

Figure. 17 rydFixr service verification at customer location

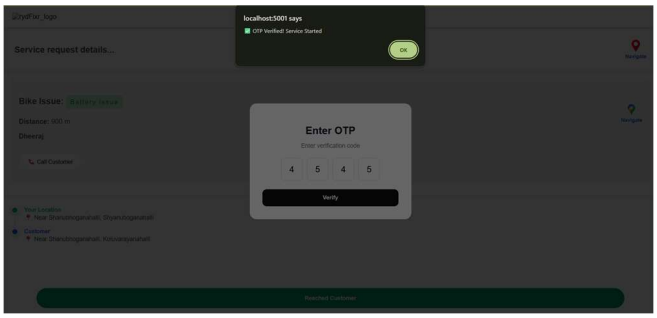

Figure. 18 rydFixr service request verified page

The rydFixr system demonstrated the efficient, reliable and precise performance during testing. Service requests which booked by customers were successfully broadcast and accepted in real time, with an average response time of around 5-8 seconds. The location-based matching, combined with dynamic radius expansion, ensured a high success rate in connecting customers with nearby Fixperts. The request locking mechanism prevented duplicate acceptances, while OTP verification provided secure service initiation. Overall, the system improved response speed, transparency, and user experience compared to traditional methods, confirming its effectiveness as a real-time bike repair assistance platform.

VI.CONCLUSION

The above discussion has been used to bring forward rydFixr which is a digital platform built on the principles of trust. The system has been created to improve efficiency and credibility associated with bike repairs. The introduction of processes like workflow structure, geo-based match, escrow payment system, and service proof has been able to tackle the issues associated with conventional methods of bike repairs. The system provides control and assurance for each of the service process steps ensuring fewer disputes between clients and fixers. Other features like tracking and warranty through rework have been provided to further ensure quality services.

VII. AUTHOR'S CONTRIBUTION

K. Guru Dheeraj Reddy is the first author and primarily responsible for the design and development of the rydFixr system, including frontend implementation, backend implementation, real-time communication integration, and overall system workflow. He also conducted testing, analyzed results, debugged the rydFixr end-end. Rohith D Raj contributed by collecting and organizing background information, reviewing reference papers, and supporting the documentation process. He assisted in structuring the literature-related content for the study. Dr. Wasim Yasin provided academic supervision and guidance throughout the project. He contributed to refining the methodology, validating the system design, and reviewing the final research paper for quality and clarity.

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